离散数学朋辈辅学

Lec 4 - Chap 9

廖謜余
Exodddus
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- **binary relation**
- the graph of function
- relation on a set
  - How many binary relations are there on a set A with n elements?
  - 连接矩阵共 n^2 个位置，每个位置可取 0 或 1，共 $\mathbf{2^{n^2}}$ 个

- reflexive
  - $\forall a \in A,\ (a,a) \in R$. 矩阵对角线全为1
- irreflexive
  - $\forall a \in A,\ (a,a) \notin R$. 矩阵对角线全为0
  - 可以既非 reflexive 又非 irreflexive 吗?
- symmetric
  - $(a,b) \in R \Rightarrow (b,a) \in R$, 对称矩阵
- antisymmetric
  - $(a,b) \in R \wedge (b,a) \in R \Rightarrow a = b$
  - $a \neq b \Rightarrow$ 不能同时有 (a,b) 与 (b,a)
  - 可以既 symmetric 又 antisymmetric 吗
  - 可以既非 symmetric 又非 antisymmetric 吗?

$$\begin{pmatrix} \mathbf{1} & 1 & 0 \\ 0 & \mathbf{0} & 0 \\ 0 & 0 & \mathbf{0} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

- How many relations are there on a set with n elements that are
  (1) reflexive?

$$M_R = \begin{pmatrix} \mathbf{1} & * & * \\ & \mathbf{1} & * \\ & * & \mathbf{1} \end{pmatrix} \quad \Rightarrow \quad \boxed{2^{n^2-n}}$$

  (2) symmetric?

$$M_R = \begin{pmatrix} * & a & b \\ a & * & c \\ b & c & * \end{pmatrix} \quad \Rightarrow \quad \boxed{2^{\frac{n(n+1)}{2}}}$$

  (3) antisymmetric?
  对角线 2 种，每对 (m_{ij}, m_{ji})（$i < j$）有 3 种：$(0,0), (1,0), (0,1)$

$$\Rightarrow \quad \boxed{2^n \cdot 3^{\frac{n(n-1)}{2}}}$$

- What about:
  (4) reflexive and symmetric?
  对角线固定为 1，上三角 $\frac{n(n-1)}{2}$ 是自由的

  $$\Rightarrow \quad \boxed{2^{\frac{n(n-1)}{2}}}$$

  (5) transitive?
  idk :)

4) The properties of relation operations

Suppose that R, S are the relations from A to B, T is the relation from B to C, P is the relation from C to D, then

(1) $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$

(2) $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$

(3) $(\overline{R})^{-1} = \overline{R^{-1}}$

(4) $(R - S)^{-1} = R^{-1} - S^{-1}$

(5) $(A \times B)^{-1} = B \times A$

(6) $\overline{R} = A \times B - R$

(7) $(S \circ T)^{-1} = T^{-1} \circ S^{-1}$

(8) $(R \circ T) \circ P = R \circ (T \circ P)$

(9) $(R \cup S) \circ T = R \circ T \cup S \circ T$

- reflexive closure
  - $r(R) = R \cup I_A$
- symmetric closure
  - $r(R) = R \cup R^{-1}$
- transitive closure
  - connectivity relation R^*
  - $t(R) = R^*$
  - 如何证明以上公式? ① $R \subseteq R^*$ ②R^* is transitive ③在所有满足以上条件的集合中，R^*是最小的。

① containing R

② is a transitive relation

Suppose (a, c) and (c, b) are in R^*. Show that (a, b) is in R^*.

③ is the smallest transitive relation which contains R

Now suppose that S is any transitive relation which contains R. We must prove S contains R^* to show R^* is the smallest relation.

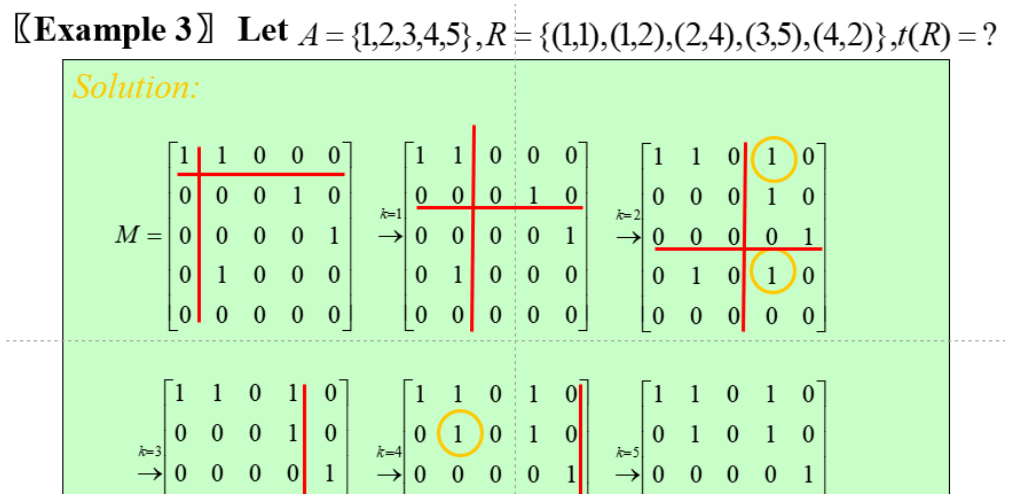
朴素算法 $O(n^4)$

```
result := M_R;  power := M_R
for i := 2 to n
  power := power ⊙ M_R    // 布尔矩阵乘法
  result := result ∨ power
// result = M_{R*}
```

Warshall 算法 $O(n^3)$

```
W := M_R
for k := 1 to n
  for i := 1 to n
    for j := 1 to n
      w[i][j] := w[i][j] ∨ (w[i][k] ∧ w[k][j])
// W = M_{R*}
```

- 一种快速得到 R^* 的方法：

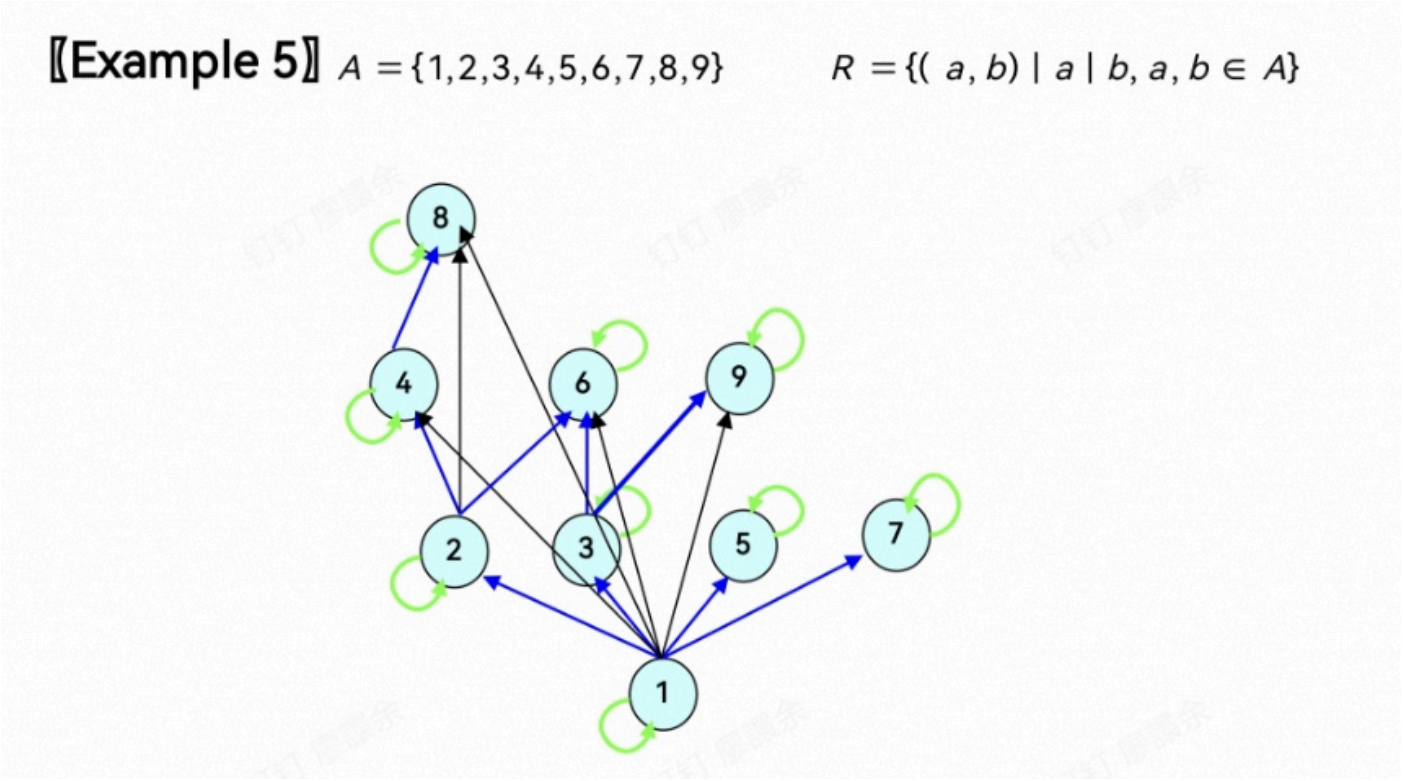


- Relation R is an equivalence relation if R is
  - reflexive
  - symmetric
  - transitive
- 若 R_1, R_2 是集合 A 上的等价关系，则 $R_1 \cup R_2$ 是 A 上的 **reflexive** 且 **symmetric** 的关系。
  证明:
  (1) reflexive: $\forall a \in A,\ \because\ (a,a) \in R_1,\ (a,a) \in R_2,\ \therefore\ (a,a) \in R_1 \cup R_2$ √
  (2) symmetric: $(a,b) \in R_1 \cup R_2$
  $\Rightarrow\ (a,b) \in R_1$ 或 $(a,b) \in R_2$
  $\Rightarrow\ (b,a) \in R_1$ 或 $(b,a) \in R_2\ \Rightarrow\ (b,a) \in R_1 \cup R_2$ √
  transitive?

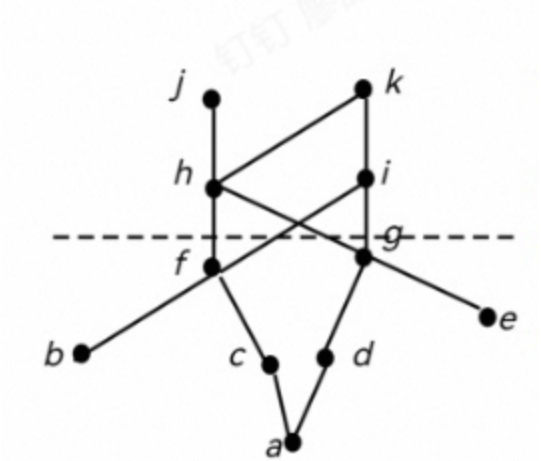
9.6 Partial Ordering

- partial ordering
  - reflexive
  - antisymmetric
  - transitive
- comparable & incomparable
- total order
  - $(\mathbb{R}, \leq)$ 是 total order
  - $(\mathbb{Z}^+, |)$ 是 partial order，但不是 total order
  - $(\mathcal{P}(A), \subseteq)$ 是 partial order，一般也不是 total order
- lexicographical order
  - 从左到右逐位比较，遇到第一处不同，就按这一位的大小决定整体顺序

- chain / antichain
  - 若 $B \subseteq A$ 且 $(B, \preceq)$ 是 total order，则 B 称为 $(A, \preceq)$ 的一个 chain
  - 若 $B \subseteq A$ 且任意不同的 $a, b \in B$ 都 incomparable，则 B 称为 $(A, \preceq)$ 的一个 antichain
- Hasse Diagram



- maximal / minimal element
- greatest / least element
- upper / lower bounds
- least upper bounds / greatest lower bounds



- maximal elements：j, k
- minimal elements：a, b, e
- greatest element：不存在
- least element：不存在
- 对 $\{c, d\}$ 而言，a 是一个 lower bound，并且是 greatest lower bound
- 对 $\{f, g\}$ 而言，h, i, j, k 都是 upper bounds，但没有唯一的 least upper bound

- well-ordered set
  - 每个非空子集都有 least element 的 total order，相当于确保有左端点的 total order.
- lattice
  - 其中任意两个元素都有 least upper bound 和 greatest lower bound 的 poset。
- topological sorting